

BioMed Capabilities Experience Builder

Review Findings, Narrative Requirements, and Implementation Specification

Prepared for: Jeff Franzen

Prepared from: Troy Rathbun follow-up email, Jennifer review feedback, full call narrative, and subsequent design clarification

Working use: Desktop build companion, Obsidian note, and PDF follow-along document

Document intent: Preserve the full reasoning, not compress the requirements

Table of Contents

1. Purpose of This Document
2. Source Material Used
3. Overall Review Interpretation
4. Primary Design Philosophy
5. Application-Wide Requirements
6. Home Page Requirement
7. Chapter Section and Nine-Tile Navigation
8. Tile 1 - Blood 101
9. Tile 2 - Blood Journey
10. Tile 3 - Hospital Distribution
11. Tile 4 - Future Demand
12. Tile 5 - BioMed Collections Overview
13. BioMed Collections - Page Structure
14. BioMed Collections - Mobile Collections Section
15. BioMed Collections - Fixed Sites Section
16. BioMed Collections - Diversity and Population Representation
17. BioMed Collections - Volunteer Section
18. Content Realignment Summary
19. Tile 6 - Jurisdiction Dashboard
20. Tile 7 - BioMed Operations Workbench
21. GOAT Hierarchy and Boundary Logic
22. Tile 8 - Hospital Network
23. Tile 9 - Explore Regions Overview
24. Explore Regions - Regional Overview
25. Explore Regions - Fixed Site Selection
26. Fixed Site Trade Area Layer
27. Fixed Site Infographic Concept
28. Red Cell and Platelet Trade Area Roadmap
29. Sickle Cell Content in Explore Regions
30. Data Sources & Methodology Modal
31. Modal Accordion Structure

32. Data Source Inventory Requirements
 33. KPI and Metric Standards
 34. Marketing and Communications Review
 35. User Experience and Interaction Standards
 36. Implementation Sequence
 37. Dependencies and Items Needed from Troy
 38. Open Questions
 39. Quality Assurance Checklist
 40. Future Project - Real-Time Cancellation Dashboard
 41. Final Interpretation of the Review
 42. One-Page Builder View
 43. Suggested Review Script for the Next Meeting
 44. Closing Build Principle
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1. Purpose of This Document

This document is the single working specification for the next round of changes to the BioMed Capabilities Experience Builder. It is not intended to be a short punch list, and it is not intended to be a generic punch list. It is a narrative requirements document that captures the written email, the phone discussion, the reviewer reactions, the implied design intent, and the implementation direction that should guide the build.

The central point of the review was that the existing experience is strong. The requested work is not a rejection of the current build. The feedback indicates that the current design has already created confidence, interest, and excitement. The next step is to sharpen the experience so it tells a more complete BioMed story, carries stronger source credibility, shows the role of volunteers, supports regional storytelling, and allows leadership to understand local impact as well as national scale.

This document should be kept open while editing the Experience Builder application. It is written to answer four questions at every stage of the build:

1. What exactly needs to change?
2. Why does it need to change?
3. What was the discussion behind the change?
4. What implementation choices should be made so the final product matches the intent of the review?

The original email gives the formal list of changes. The call transcript gives the meaning behind those changes. Both matter. For example, the email says to add an Explore Regions tile. The call explains that Explore Regions is not just a new map tile. It is intended to become the local story, the community impact story, the diversity story, and potentially the fixed-site trade-area story. Similarly, the email says to verify data accuracy and sources. The call makes clear that this is not a minor administrative task. The very first concern raised was source traceability. That means data credibility needs to become part of the architecture of the application.

This document therefore treats each requirement as more than a task. It treats each requirement as a design decision that should remain visible to the builder, Troy, Jennifer, and any later stakeholder who asks why the experience was structured this way.

2. Source Material Used

This specification is based on three streams of information.

The first stream is Troy Rathbun's follow-up email. That email provides the formal notes from the review and contains the most authoritative list of requested changes. It identifies the fixed-site links, the revised homepage language, the desired nine-tile structure, the tile-by-tile enhancement requests, the content movement requests, the operations workbench requirements, and the Explore Regions requirements.

The second stream is the call transcript. The transcript provides the context behind the email. It captures comments such as Jennifer liking the current page, Troy noting that several changes are easy structural changes, the desire to create stronger regional storytelling, the uncertainty around some Future Demand numbers, the excitement around fixed-site trade-area polygons, the need to recognize boundary exceptions, and the future idea of a real-time cancellation dashboard.

The third stream is the follow-up clarification that the data-source capability should be implemented as a pop-up modal with accordion sections. This is a critical architectural choice. It should not appear in the application as a visible tenth tile. It should be available as an application-wide credibility layer, likely through an "About the Data" link or information icon.

The formal fixed-site links provided in the email are:

FS Map Layer:

<https://arc-nhq-gis.maps.arcgis.com/home/item.html?id=6af8a323a0b5468f9427baa4d6ee7481#visualize>

Red Cell Fixed Sites Trade Area Dashboard:

<https://arc-nhq-gis.maps.arcgis.com/apps/dashboards/f2e70959109342fca8da1fd33cfc54b2>

Troy also indicated that additional information, dashboards, KPIs, and layer access will be provided later. Those future inputs are documented in the dependency section of this specification.

3. Overall Review Interpretation

The overall review was positive. The repeated tone of the conversation was that Jennifer liked the experience, Troy was pleased with the direction, and the current design had enough value that the discussion moved immediately into refinement rather than rework.

Several important conclusions can be drawn from that tone.

First, the experience should not be visually rebuilt from scratch. The existing visual approach, especially in the Operations Workbench, was received well. The work should focus on improving the flow, relocating content into better narrative homes, adding missing storylines, and connecting the existing design to better data governance.

Second, the reviewers are thinking about the audience. This is not only a GIS tool. It is not only a dashboard. It is not only a story map. The audience includes people who may not deeply understand BioMed. Some users will need introductory content. Some will need national context. Some will want to understand what happens in their region. Some will challenge the numbers. Some will ask how BioMed boundaries align with Humanitarian Services or other Red Cross structures. The application needs to serve all of those audiences without becoming cluttered.

Third, the experience needs to tell a more human story. Volunteers, donor diversity, sickle cell, community impact, local hospitals, and fixed-site reach all came up because the current application is strong operationally but can become even stronger when it shows how BioMed connects to people and communities.

Fourth, the data-source question is central. The review began with the question of whether there is a list showing where the statistical data came from. That is a signal that source transparency will become a recurring need. If the application is shown to executives or national stakeholders, every number must be defensible. The correct response is not to hide the numbers or make the experience less data-driven. The correct response is to add a source-traceability layer that makes the application more trustworthy.

Fifth, Explore Regions is the largest new storytelling opportunity. It should not be built as an afterthought. It should become the place where all the local context comes together: chapters, hospitals, collections, products, staging locations, donor populations, diversity, sickle cell, fixed sites, and trade areas.

4. Primary Design Philosophy

The revised application should be built around a simple but powerful progression: understand BioMed, understand the journey, understand the network, understand the future, understand collections, understand geography, understand operations, understand hospitals, and finally understand regional impact.

The first part of the application should educate. The middle should orient the user geographically and operationally. The final part should localize the story.

This progression matters because users rarely arrive with the same level of background knowledge. Some will know blood services deeply. Others may be seeing the BioMed ecosystem for the first time. If the experience jumps immediately into fixed-site trade areas or operational hierarchy, it may lose users who need a foundation. If it remains only high-level, it will fail users who want regional or operational detail. The nine-tile structure should solve that by creating a deliberate pathway.

The application should avoid feeling like a collection of unrelated dashboards. It should feel like a guided architecture. Each tile has a purpose. Each tile prepares the user for the next tile. Each tile should make the following tile easier to understand.

The design should also recognize that the application has two modes of value. One mode is presentation value. Leadership can open it, see a polished story, and understand the organization at a high level. The second mode is working value. Analysts, planners, and managers can use it to answer specific questions about boundaries, regions, fixed sites, infrastructure, and hospitals. The best version of the application serves both modes without overloading either one.

5. Application-Wide Requirements

Several requirements cut across the entire application and should not be treated as tile-specific items.

The first application-wide requirement is source transparency. Every meaningful statistic should be traceable to a documented source. This requirement should be implemented through the Data Sources & Methodology modal described later in this document.

The second application-wide requirement is geographic clarity. Wherever the application uses division, region, district, chapter, or other organizational geography, it should be clear what hierarchy is being used and whether it aligns with Humanitarian Services or other Red Cross boundary structures. Boundary exceptions should not be hidden. They should be explained simply and calmly.

The third application-wide requirement is consistent filtering where appropriate. The email specifically calls out Division, Region, and Chapter filters for multiple sections. The call also discussed district and the GOAT taxonomy. The application should use consistent language and consistent behavior so users do not have to relearn filters in every tile.

The fourth application-wide requirement is volunteer visibility. Volunteers should not appear once and disappear. The review called for volunteer sections in Blood 101, Blood Journey, Hospital Distribution, and Collections. The deeper design intent is that volunteers should be recognized as part of the operational and human story of BioMed.

The fifth application-wide requirement is local storytelling. Explore Regions is the primary home for this, but the idea should influence the whole application. Users should be able to move from national scale to local meaning.

The sixth application-wide requirement is restraint. The reviewers liked the current work. The build should not become visually chaotic. Additions should be clean, modular, and purposeful. The goal is to strengthen the existing experience, not bury it under every possible data point.

6. Home Page Requirement

The homepage language must change from:

“Explore the Experience”

To:

“Inside BioMed Capabilities”

This may appear like a small text change, but it carries important positioning. “Explore the Experience” sounds like an invitation to browse. “Inside BioMed Capabilities” sounds like an authoritative operational view. Jennifer specifically recommended this type of change because the application should speak more broadly to BioMed and not merely invite a user into a generic experience.

The homepage should establish that this application is about capability, scale, operations, and impact. It should feel useful for an executive, a planner, a communicator, and a regional leader.

Implementation guidance:

- Use the exact phrase “Inside BioMed Capabilities” unless Troy or Jennifer later requests a different final label.
 - Keep the surrounding homepage design mostly intact if it is already visually working.
 - Avoid adding too much explanatory text to the homepage. The title change should sharpen the framing without making the landing page dense.
 - Consider adding a subtle “About the Data” control somewhere persistent, but not in a way that competes with the main title.
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7. Chapter Section and Nine-Tile Navigation

The chapter section should be reorganized into a final nine-tile structure. This structure is a major architectural requirement because it determines how the user moves through the BioMed story.

The final tile structure is:

1. Blood 101
2. Blood Journey
3. Hospital Distribution
4. Future Demand
5. BioMed Collections
6. Jurisdiction Dashboard
7. BioMed Operations Workbench
8. Hospital Network
9. Explore Regions

The written email indicates that the early tiles are informational in nature. That means these pages should not be overloaded with dashboards or advanced interaction. They should educate, orient, and guide. Later tiles can carry heavier mapping and drill-down functionality.

The Biomed Blood Map and Jurisdiction Dashboard should be merged into the Jurisdiction Dashboard tile. The merged tile should become the authoritative geography reference rather than splitting the user across two similar map concepts.

The new Explore Regions tile should be added as tile nine. This tile is not simply a replacement for removed content. It is the new home for the regional story, community impact, sickle cell context, diversity representation, and fixed-site trade-area exploration.

Implementation guidance:

- Keep the total visible tiles at nine.
- Do not add a visible tenth tile for data sources. The data-source function should be a modal, not a navigation tile.
- Ensure tile names are concise and consistent.
- Ensure the visual order reinforces the story from foundational to regional.
- If tile cards include subtitles, use those subtitles to clarify the purpose of each tile without creating long blocks of text.

Potential tile subtitles:

- Blood 101: "Foundations of blood services"
 - Blood Journey: "From donor engagement to patient impact"
 - Hospital Distribution: "How products move to hospitals"
 - Future Demand: "Projected needs and planning signals"
 - BioMed Collections: "Mobile collections, fixed sites, and donor reach"
 - Jurisdiction Dashboard: "Boundaries, territories, and operational geography"
 - BioMed Operations Workbench: "Infrastructure and operational support network"
 - Hospital Network: "Hospital relationships and distribution reach"
 - Explore Regions: "Local story, community impact, and fixed-site reach"
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8. Tile 1 - Blood 101

Blood 101 was received well. The review did not call for major restructuring of this tile. Jennifer liked the page, and the only major requested enhancement was the addition of volunteer content.

The purpose of Blood 101 is to provide foundational understanding. It should help users who are not deep BioMed experts understand the basics before they move into the donor journey, hospital distribution, demand, collections, and regional impact.

The key requirement is to add a volunteer section. This should not be treated as a token mention. The review made clear that volunteers need to be visible as part of the BioMed story. Volunteers are not peripheral to the mission. They help make the mission work. Blood 101 is a good place to introduce that idea because this tile explains the foundation of the operation.

The volunteer section in Blood 101 should answer basic questions:

- Where do volunteers fit into the blood services ecosystem?
- What kinds of roles do they play?
- How do they support donors, staff, and communities?
- Why does their contribution matter to BioMed capability?

The tone should be clear, respectful, and mission-oriented. It should avoid overclaiming or using language that Marketing and Communications may later need to soften. The email specifically says to validate content with Marketing & Communications. That means all volunteer narrative should be drafted in a way that is accurate but flexible.

Recommended section structure:

1. Short introductory paragraph explaining that volunteers support the blood mission across community engagement and donor-facing activities.
2. Three to five short callouts showing where volunteers contribute.
3. A small visual or icon row if space allows.
4. Optional source note if any volunteer counts or volunteer-hour metrics are used.

Potential language to use as a placeholder:

“Volunteers help extend the reach of the blood mission by supporting donors, communities, and operational teams. Their work helps create a welcoming experience, supports local engagement, and strengthens the connection between BioMed operations and the communities served.”

Do not add detailed volunteer statistics unless the source is documented. If statistics are added, they must be connected to the Data Sources & Methodology modal.

9. Tile 2 - Blood Journey

Blood Journey was also received positively. The tile should remain focused on the donor-to-patient story, but it needs to absorb content that currently lives elsewhere and logically belongs here.

The call and email identify three items that should move into Blood Journey:

- Clara AI
- RapidPass

- Track Your Blood

These items should move from the current Page 6 location into Blood Journey because they relate to the donor experience and the broader blood journey. They are not primarily collections-infrastructure content. They explain how the donor interacts with the system, how the experience is improved, and how the journey becomes more transparent or efficient.

The movement of this content should improve the story. Blood Journey should become the place where the user sees the complete experience from donor engagement to blood movement to patient impact. Clara AI, RapidPass, and Track Your Blood are all journey touchpoints. They should be framed as part of the system that supports donors and strengthens the connection between donation and outcome.

A volunteer section should also be added here. This volunteer section should not repeat the Blood 101 volunteer section word-for-word. In Blood 101, the volunteer story is foundational. In Blood Journey, the volunteer story should be experiential. It should show how volunteers support the donor journey, create human connection, assist with donor flow, and reinforce the mission at the point of community contact.

Recommended Blood Journey structure:

1. Overview of the donor-to-patient journey.
2. Key stages of the journey.
3. Technology and engagement tools: Clara AI, RapidPass, Track Your Blood.
4. Volunteer role in the journey.
5. Connection to patient impact.

Implementation guidance:

- Avoid making the moved technology content feel bolted on. It should be woven into the journey narrative.
- If Clara AI, RapidPass, or Track Your Blood have existing visuals, keep them but reposition them under the Blood Journey tile.
- Use labels that explain why each tool matters to the donor or to the journey.
- If the content has metrics, make sure those metrics are source-documented.

Potential placeholder text:

“The blood journey is supported by both human connection and enabling technology. Tools such as RapidPass, Clara AI, and Track Your Blood help streamline the donor experience, support engagement, and help donors understand the path from donation to impact. Volunteers remain an important part of that journey, helping create a welcoming and mission-centered experience for donors.”

10. Tile 3 - Hospital Distribution

Hospital Distribution remains largely approved. The requested enhancement is the addition of volunteer content. The tile should continue to explain how blood products move to hospitals and how distribution supports patient care.

The call transcript originally referred to this tile in a way that may have sounded like “Costco Distribution” because of transcription error. The email confirms the intended tile is Hospital Distribution. The final document and build should use Hospital Distribution.

The volunteer story here should be different from the volunteer story in Blood 101 and Blood Journey. In Hospital Distribution, the focus should be on how volunteers support the broader chain of mission delivery. Depending on the actual volunteer roles and what Marketing approves, this may include logistics support, community support, transportation-related support, or mission enablement.

The key design challenge is to keep the tile focused on hospital distribution while still recognizing volunteers. The section should not distract from the distribution story. It should show that volunteers are part of the support structure that allows the mission to connect donors, blood products, hospitals, and patients.

Recommended section structure:

1. Overview of hospital distribution.
2. Distribution network explanation.
3. How products reach hospitals.
4. Volunteer contribution to mission delivery.
5. Link or transition to Hospital Network tile if appropriate.

Potential placeholder language:

“Hospital distribution connects collections and processing to patient care. Volunteers and staff together support the mission infrastructure that helps blood products reach communities and hospitals. Where appropriate, volunteer support should be shown as part of the larger delivery ecosystem.”

Implementation guidance:

- Use volunteer content that has been validated by Marketing & Communications.
 - Avoid implying volunteers perform functions they do not perform.
 - Keep this section concise and mission-focused.
 - If there are distribution metrics, connect them to the data-source modal.
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11. Tile 4 - Future Demand

Future Demand needs source validation. The review did not indicate that the section is poorly designed. The concern was about confidence in the numbers. Troy said Jennifer was surprised by some of the numbers and wanted to double-check the decks and data to make sure they were accurate.

That is an important distinction. The task is not to remove the numbers. The task is to verify them, document them, and make them defensible.

Future Demand is likely to be one of the most scrutinized tiles because forecasted demand affects strategy, planning, investment, and executive judgment. Any chart or KPI in this section must be traceable to a source. If a number is derived, the calculation should be documented. If a projection is based on an assumption, the assumption should be named.

Required actions:

- Review every number shown in Future Demand.
- Identify the source of each number.
- Confirm whether each number comes directly from a source or is calculated.
- Document the calculation method for calculated values.
- Confirm whether the number is national, regional, chapter-level, or some other geography.
- Confirm refresh date or effective date.

- Add entries to the Data Sources & Methodology modal.
- Decide whether any caveat is needed in the tile itself.

The Future Demand tile should remain visually compelling, but it must become source-backed. If reviewers are surprised by numbers, the application should be able to answer the question “Where did that come from?” immediately.

Implementation guidance:

- Add a small “About this data” link or information icon near Future Demand charts. This should open the application-wide Data Sources & Methodology modal, ideally with the Future Demand accordion section expanded.
- If the modal cannot open to a specific accordion section, the modal should still include a clearly labeled Future Demand data source entry.
- Avoid cluttering the tile with long footnotes. Use the modal for detail.
- Use concise chart captions to explain what each number represents.

Potential caption style:

“Projected demand based on source dataset and methodology documented in About the Data.”

Do not finalize the Future Demand tile until the source list has been assembled and Troy has confirmed that the numbers match the intended source decks or datasets.

12. Tile 5 - BioMed Collections Overview

BioMed Collections is one of the most important tiles in the revised structure. The email says it should be primarily informational with high-level supporting data. The call expands this by explaining that Jennifer wants the section to help users understand mobile collections, fixed sites, territory footprint, and the role of volunteers and diversity.

The purpose of this tile is not to become the deepest operational dashboard in the application. Its purpose is to educate users on how collections work and to provide enough high-level data for context.

This tile should answer:

- What are BioMed collections?
- How do mobile collections work?
- How do fixed sites work?
- How do mobile and fixed strategies differ?
- How are collections organized geographically?
- How do division, region, and chapter filters help interpret collection data?
- How do volunteers support collections?
- How do diversity and population representation connect to collections?

The tile should include territory footprint map visualizations. These maps do not need to be overly complex, but they should help the user understand geography. The review specifically mentions filters for Division, Region, and Chapter. These should be available where they make sense.

The tile should also include a boundary disclaimer. This disclaimer is important because BioMed and Humanitarian Services boundaries do not always align. If a user compares BioMed geography with HS or DCS geography and sees differences, the application should already have explained that exceptions exist.

Recommended disclaimer language:

“BioMed operational boundaries do not always align one-to-one with Humanitarian Services boundaries. Some geographic and operational exceptions exist, including areas where chapter, region, or division relationships may differ.”

If space is limited, use a shorter version:

“Boundary note: BioMed operational geography may not always align exactly with Humanitarian Services boundaries.”

The call specifically mentioned California as an area where exceptions may be important. If there are known exception areas, the modal or a tooltip can include more detail. The main tile should not become bogged down with boundary explanations, but it should prevent confusion.

13. BioMed Collections - Page Structure

The email gives a clear internal structure for the BioMed Collections tile.

Page 2 should become Mobile Collections.

Page 3 should become Fixed Sites.

The existing Page 6 content should move into Blood Journey.

The existing Page 7 Sickle Cell content should move into Explore Regions.

New sections should include Diversity and Volunteers.

This structure should be implemented as a deliberate content reorganization, not as a random shuffle. The goal is to put each topic in the place where it best supports the story.

Mobile Collections belongs in BioMed Collections because it is a core collection model.

Fixed Sites belongs in BioMed Collections because it is a core collection model and part of the growth strategy.

Clara AI, RapidPass, and Track Your Blood belong in Blood Journey because they are donor journey tools.

Sickle Cell belongs in Explore Regions because it is better connected to community impact, diversity, population representation, and local story.

Diversity belongs in Collections and Explore Regions because donor populations and community representation directly affect the collection story and the regional impact story.

Volunteers belong throughout because they support the mission across multiple points in the journey.

Implementation guidance:

- Rebuild the internal navigation of BioMed Collections so the sections have a clear order.
- Use consistent section naming.

- Avoid duplicate content between Mobile Collections and Fixed Sites.
- Use high-level metrics only unless deeper drill-down is required.
- Push more detailed regional fixed-site analysis into Explore Regions.

Recommended BioMed Collections internal sequence:

1. Collections overview.
 2. Mobile Collections.
 3. Fixed Sites.
 4. Diversity and population representation.
 5. Volunteer contribution in collections.
 6. Notes on geography and boundary alignment.
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14. BioMed Collections - Mobile Collections Section

The Mobile Collections section should be added to Page 2. It should explain mobile collections at a high level and provide national context with filters.

The section should include:

- National overview.
- Division filter.
- Region filter.
- Chapter filter.
- Possibly a high-level map or territory footprint visualization.

The purpose is to explain the role of mobile collections in the broader BioMed model. Mobile collections are likely the way many users think about blood donation because they connect to community drives, sponsors, and local events. The section should make that clear without becoming too detailed.

The national overview should set the scale. The filters should allow users to quickly understand how the mobile collection model appears in different parts of the country.

Potential metrics, depending on data availability:

- Number of mobile drives.
- Collections from mobile drives.
- Sponsors or sponsor locations.
- Donor activity.
- Mobile collection contribution compared with fixed-site contribution.
- Regional or chapter variation.

Implementation guidance:

- Use a map if it helps orientation, not merely because a map is available.
- If the map is used, make it simple enough to read at national and filtered views.
- Include clear labels for Division, Region, and Chapter filters.
- If filters cannot all be implemented immediately, prioritize Region and Division first, then Chapter.
- Avoid unsupported KPIs until Troy provides the additional metrics he mentioned in the call.

Narrative guidance:

The section should explain that mobile collections extend BioMed's reach into communities through sponsor sites, community partnerships, and local access points. It should connect mobile collections to mission reach rather than showing them as only operational events.

Potential placeholder language:

"Mobile collections extend the blood mission into communities by bringing donation opportunities closer to donors, sponsors, and local partners. This section provides a national view of the mobile collection model and allows users to filter by division, region, and chapter to understand how mobile collections support different geographies."

15. BioMed Collections - Fixed Sites Section

The Fixed Sites section should be added to Page 3. Jennifer specifically wanted to understand fixed sites and the Fixed Site Growth Program. This section should explain what fixed sites are, why they matter, and how they differ from mobile collections.

The section should include:

- Overview and description of fixed sites.
- Fixed Site Growth Program.
- National overview.
- Division filter.
- Region filter.
- Chapter filter.
- Number of fixed sites.
- Additional fixed-site collection data.
- Possibly a high-level map visualization.

This section should be high-level. The deeper fixed-site trade-area drill-down should live in Explore Regions. BioMed Collections should explain the strategy. Explore Regions should show the local impact.

The Fixed Site Growth Program should be treated as a strategic initiative. The section should not simply say that fixed sites exist. It should explain why growth matters. Fixed sites may support capacity, reliability, donor engagement, operational planning, and regional collection strategy. Exact claims should be validated with Troy and Marketing/Communications before final publication.

Potential metrics:

- Total fixed-site count.
- Fixed sites by division.
- Fixed sites by region.
- Collections by fixed site.
- Donor visits.
- Collection trends.
- Fixed-site contribution to overall collections.

Implementation guidance:

- Use the FS Map Layer as a source or visual reference.

- Confirm whether the existing layer is authoritative and current.
- Connect fixed-site metrics to documented data sources.
- Include a short map if it adds clarity.
- Keep the section educational; push complex trade-area analysis to Explore Regions.

Potential placeholder language:

“Fixed sites provide standing locations for blood donation and support a strategic collection model alongside mobile collections. The Fixed Site Growth Program should be presented as part of BioMed’s broader effort to strengthen access, capacity, and community reach. National and filtered views should help users understand the distribution of fixed sites across divisions, regions, and chapters.”

16. BioMed Collections - Diversity and Population Representation

The email requests a new Diversity section and identifies population representation for African American, Latino, and LGBTQ+ communities. The call also connected diversity to sickle cell and regional/community impact.

This content should be handled carefully and clearly. The goal is not to make a broad social statement. The goal is to explain how population representation, donor diversity, and community engagement matter to the blood mission.

The Diversity section in BioMed Collections should remain high-level. It should introduce the concept that donor populations and community representation matter. Explore Regions can then localize that story.

Topics may include:

- African American donor representation.
- Latino donor representation.
- LGBTQ+ donor engagement.
- Population representation in local markets.
- Connection between community demographics and donor strategy.
- Relationship to sickle cell needs, where appropriate and approved.

Implementation guidance:

- Use source-backed data only.
- Avoid unsupported statements.
- Validate language with Marketing & Communications.
- Avoid making the section too dense.
- Use the Data Sources & Methodology modal for definitions and source details.
- Consider a neutral title such as “Donor Diversity and Population Representation” rather than simply “Diversity” if that better fits the audience.

Potential placeholder language:

“Population representation helps BioMed understand how donor engagement aligns with the communities served. This section introduces high-level context for donor diversity, including African American, Latino, and LGBTQ+ populations, and provides a foundation for the more localized community impact story in Explore Regions.”

This section should not replace the Sickle Cell content in Explore Regions. Instead, it should introduce the broader theme. Explore Regions should carry the deeper local story.

17. BioMed Collections - Volunteer Section

The email explicitly adds volunteers to BioMed Collections. The call also reinforced that volunteers should be shown across multiple tiles.

The Collections volunteer section should focus on how volunteers support collection operations and donor engagement. It should be distinct from the Blood 101 and Blood Journey volunteer sections.

Potential angles:

- Volunteers help support donor-facing activity.
- Volunteers help connect communities to the blood mission.
- Volunteers strengthen local engagement.
- Volunteers help create a welcoming environment.
- Volunteers contribute to the overall collection ecosystem.

Implementation guidance:

- Confirm specific volunteer roles before making operational claims.
- Use approved language from Marketing & Communications.
- Use metrics only if source-backed.
- Consider a small visual card titled “Volunteer Contribution” or “Volunteers in Collections.”

Potential placeholder language:

“Volunteers support the collection mission by strengthening community connection and helping create a positive donor experience. Their role should be visible as part of the broader BioMed collection ecosystem.”

18. Content Realignment Summary

Several content moves were requested because the current locations do not fully match the story being told.

Move Page 6 content into Blood Journey:

- Clara AI
- RapidPass
- Track Your Blood

Reason: These are donor journey and engagement tools. They belong with the donor-to-patient narrative rather than inside the collections tile.

Move Page 7 content into Explore Regions:

- Sickle Cell

Reason: Sickle Cell is better understood as part of local community impact, diversity, donor representation, and patient need. It is not only a collection operations topic. Explore Regions provides the proper context for connecting Sickle Cell to local demographics, community outreach, and regional impact.

These moves should be handled carefully. Content should not be copied in a way that creates duplicate outdated versions. The original sections should either be removed from their old locations or converted into light references that link to the new home.

Implementation guidance:

- Move, do not duplicate, unless there is a clear reason to cross-reference.
 - Update navigation labels and internal links after the move.
 - Review all captions and headers after relocation so they match the new section context.
 - Confirm that the moved content still flows logically in its new location.
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19. Tile 6 - Jurisdiction Dashboard

The Biomed Blood Map and Jurisdiction Dashboard should be merged into a single Jurisdiction Dashboard tile. This merged tile should become the authoritative geographic reference for operational boundaries, territories, and related geography.

The email asks whether there are recommendations for this tile. The call suggests that the existing jurisdiction dashboard is acceptable as-is, but there may be opportunities to make it more useful and efficient.

The core purpose of this tile should be geographic orientation. Users should understand the footprint, boundaries, and jurisdictional structure. If the boundaries update in the source layer, the dashboard should update accordingly. That dynamic behavior is valuable and should be preserved.

Recommended functions:

- Display BioMed operational boundaries.
- Support division, region, district, and chapter context if available.
- Explain alignment exceptions with Humanitarian Services.
- Allow users to inspect boundaries without needing advanced GIS knowledge.
- Serve as a reference point for users who later view Operations Workbench or Explore Regions.

Potential enhancements:

- Add a small boundary disclaimer.
- Add summary counts for selected region or geography.
- Add a simple legend explaining boundary types.
- Add a clear reset button.
- Add an “About these boundaries” link that opens the Data Sources & Methodology modal to the boundary section.
- Use bookmarks or simple navigation for major geographies.

Implementation guidance:

- Do not overcomplicate this tile. Its value is clarity.
- Ensure boundary layers are authoritative and connected to the appropriate live or maintained source.

- Confirm whether the merged map should retain all functionality from both original components.
- Remove redundant controls that confuse users.
- Make sure the merged tile has a clear title and purpose.

Potential title:

“Jurisdiction Dashboard”

Potential subtitle:

“Explore BioMed operational geography, boundaries, and territory relationships.”

20. Tile 7 - BioMed Operations Workbench

The Operations Workbench received especially positive feedback. Troy said Jennifer liked the layout and the ability to click layers. The review specifically requested retaining the design and enhancing it.

This is important. The build should not replace the Operations Workbench with a new concept. The current design has credibility. The correct move is to strengthen it with better hierarchy, infrastructure layers, and collapsible navigation.

The Operations Workbench should focus on infrastructure. It should answer practical questions such as:

- Where are manufacturing sites?
- Where are staging sites?
- Where are kitting sites?
- Where are IRL sites?
- Where are distribution sites?
- How does infrastructure relate to divisions, regions, districts, and chapters?
- What infrastructure exists inside a selected region?

Required infrastructure layers:

- Staging Sites
- Manufacturing Sites
- Kitting Sites
- IRL Sites
- Distribution Sites

Required filters:

- Division
- Region
- Chapter

Additional hierarchy/bookmark requirement:

- Add GOAT boundary bookmarks or filters for Division, Region, District, and Chapter.

The call clarified that GOAT is the internal taxonomy or hierarchy system used to keep BioMed geography organized. The hierarchy discussed was essentially a rollup structure where smaller operating areas roll into larger ones. The discussion mentioned collection areas rolling into districts, districts rolling into regions, and

regions rolling into divisions. Chapter boundaries should also be shown where possible, even if they do not always align perfectly with BioMed boundaries.

The side panel should be collapsible. This was explicitly discussed and considered easy to implement. A collapsible panel will give users more map space and make the application feel cleaner.

Implementation guidance:

- Retain the current visual layout.
- Add or confirm infrastructure layers.
- Confirm whether current visible data is real or placeholder.
- Replace prototype layers with authoritative layers if needed.
- Add hierarchy bookmarks for Division, Region, District, and Chapter.
- Include Chapter boundaries where possible.
- Add boundary disclaimer where hierarchy and chapter boundaries appear.
- Implement collapsible side panel for buttons or navigation.
- Ensure layer names are user-friendly.
- Avoid exposing internal jargon unless explained.

Potential disclaimer for this tile:

“BioMed hierarchy and Humanitarian Services chapter boundaries may not always align exactly. Boundary views are provided to support orientation and should be interpreted with known operational exceptions.”

21. GOAT Hierarchy and Boundary Logic

The GOAT hierarchy discussion is important because it affects more than the Operations Workbench. It affects how users interpret filters across the entire application.

During the call, the term GOAT was described as the taxonomy system used to keep everything organized. The hierarchy helps users understand how operational geographies roll up. The application should make this hierarchy usable without requiring users to understand internal data management language.

The hierarchy levels discussed include:

- Division
- Region
- District
- Chapter

The exact relationships may vary depending on BioMed and Humanitarian Services boundary structures. The application should not imply perfect one-to-one alignment where it does not exist.

Recommended implementation pattern:

- Use the hierarchy consistently across filters.
- Use plain-language labels.
- Use tooltips or modal definitions to explain terms.
- Include Chapter views where possible because users are interested in them.
- Add a disclaimer wherever Chapter boundaries are shown alongside BioMed operational boundaries.

- Use bookmarks for fast map navigation.
- Avoid making users manually zoom to major geographies if bookmarks can solve that.

Potential data-source modal entry:

Accordion: "Boundary and GOAT Hierarchy Sources"

Fields:

- Dataset name.
- Boundary level.
- Source owner.
- Effective date.
- Known alignment exceptions.
- Refresh schedule.
- Notes on BioMed versus Humanitarian Services boundary differences.

This will help prevent confusion later when someone asks why a chapter appears differently in a BioMed view than in another Red Cross operational view.

22. Tile 8 - Hospital Network

The Hospital Network tile is not ready for full build specification because Troy indicated he will send a dashboard concept later. The call suggests that this future concept may include distribution sites, distribution reach, hospitals, distribution numbers, and similar information.

Current direction:

- Keep Hospital Network as Tile 8.
- Do not overbuild it until Troy sends the concept.
- Be ready to incorporate a dashboard once the concept is provided.
- Treat Hospital Network as distinct from Hospital Distribution.

Hospital Distribution explains the product movement story. Hospital Network should likely show the network of hospitals served, geographic reach, service relationships, and distribution footprint.

Potential future content:

- Hospitals served.
- Hospital locations.
- Distribution reach.
- Products distributed.
- Distribution sites.
- Regional hospital counts.
- Service areas.
- Hospital network by division or region.

Implementation guidance:

- Hold major design decisions until Troy's dashboard concept is provided.

- If a placeholder is needed, use a simple statement that the hospital network view is under development.
 - Avoid inventing metrics before sources are confirmed.
 - Ensure any hospital data is properly sourced and governed in the Data Sources & Methodology modal.
-

23. Tile 9 - Explore Regions Overview

Explore Regions is the largest new functional and narrative requirement. It should be treated as a major feature, not a minor tile addition.

The email says Explore Regions should present the overall story of a region with emphasis on community impact. The call expands this significantly. Troy wants users to select a region, drill into regional details, understand hospitals and chapters, see collection activity and products distributed, and then select fixed sites to view trade-area polygons and supporting metrics.

The purpose of Explore Regions is to answer:

“What does BioMed mean in this local or regional context?”

This is where the application stops being only a national overview and becomes a local story.

The tile should support:

- Region selection.
- Fixed-site selection.
- Drill-down to detailed regional view.
- Community impact metrics.
- Chapters.
- Hospitals served.
- Staging locations.
- Collections activity.
- Products distributed locally.
- Additional relevant metrics to be determined.
- Sickle Cell content.
- Diversity and population representation context.
- Fixed-site trade-area polygons.

The review indicated that Jeff’s regional story map format may be useful here. Troy found a format that seemed to work well and suspected it may have been something Jeff had already built. That format should be considered a design reference for Explore Regions.

Implementation guidance:

- Start with a region selector.
- Provide an executive-friendly regional summary.
- Add a map showing the regional footprint.
- Add metrics below or beside the map.
- Add fixed-site selection only after the regional context is established.
- Use progressive disclosure. Do not show every detail at once.

- Use infographics for selected fixed sites.
 - Keep the design clean enough for leadership review.
-

24. Explore Regions - Regional Overview

The regional overview should tell the story of a selected region before the user drills into a fixed site.

Potential regional overview metrics include:

- Number of chapters.
- Hospitals served.
- Staging locations.
- Collection activity.
- Products distributed locally.
- Fixed sites.
- Mobile collection activity.
- Donor count.
- Population served.
- Diversity representation.
- Sickle Cell-related context where appropriate.

The exact metrics depend on available data. Troy said he would provide additional metrics, and those should be incorporated once received.

The regional overview should not overwhelm the user. A good design pattern would be a set of summary cards at the top, a map in the center, and supporting narrative or charts below.

Example structure:

1. Region name and short narrative summary.
2. KPI cards for chapters, hospitals, fixed sites, collections, products distributed, and staging locations.
3. Map showing regional boundary and key infrastructure.
4. Section for community impact.
5. Section for fixed-site drill-down.

Potential narrative placeholder:

“This regional view summarizes BioMed’s local footprint, including chapters, hospitals served, collection activity, staging locations, and product distribution. It is intended to help users understand how national BioMed capabilities translate into local community impact.”

Implementation guidance:

- Use the region selector as the primary control.
 - Make sure all metrics update when the region changes.
 - Show a clear empty state if data is unavailable.
 - Avoid filtering out regions silently unless stakeholders specifically approve that behavior.
-

25. Explore Regions - Fixed Site Selection

The fixed-site selection workflow is one of the most important new interactions. Troy described a desired flow where a user selects a region, sees the fixed sites in that region, selects a fixed site, and then sees the trade-area polygon and related metrics.

The flow should be:

Region selected.

Then fixed sites in that region become available.

Then user selects one fixed site.

Then the map zooms or updates to show the trade area.

Then an infographic or data panel explains what that trade area represents.

This workflow matters because it turns fixed sites from points into markets. A point tells the user where a site is. A trade area tells the user whom that site reaches.

The call mentioned that there are about 205 fixed sites. Most regions likely have at least one fixed site, but some regions may have few or possibly none. The application should handle this gracefully.

Recommended empty-state behavior:

If a selected region has no fixed sites, do not make the application appear broken. Show a message such as:

“No fixed sites are currently shown for this selected region. Regional collection and community metrics remain available above.”

There was discussion about whether to hide regions without fixed sites. The safer approach is not to hide them by default. Hiding regions may create confusion or make the application appear incomplete. Showing a clear zero-state is more transparent unless Troy specifically requests a filtered list of only fixed-site regions.

Implementation guidance:

- Filter fixed sites based on selected region.
 - Provide a fixed-site dropdown or list.
 - When a fixed site is selected, display its polygon and metrics.
 - Use map zoom, highlight, or selection styling to focus attention.
 - Keep national fixed-site distribution visible at a high level, but simplify once a region is selected.
-

26. Fixed Site Trade Area Layer

Troy provided the FS Map Layer and the Red Cell Fixed Sites Trade Area Dashboard. The call explained that the trade-area polygons were created based on donor behavior and migration patterns. This is a major storytelling advantage.

A fixed-site point is useful, but a trade-area polygon is much more powerful because it shows the real geographic reach of a fixed site. It can help users understand where donors come from, how far they travel, what population is represented, and what local market the site serves.

The fixed-site layer should support the Explore Regions tile and possibly inform the Fixed Sites section in BioMed Collections.

Layer and dashboard references:

FS Map Layer:

<https://arc-nhq-gis.maps.arcgis.com/home/item.html?id=6af8a323a0b5468f9427baa4d6ee7481#visualize>

Red Cell Fixed Sites Trade Area Dashboard:

<https://arc-nhq-gis.maps.arcgis.com/apps/dashboards/f2e70959109342fca8da1fd33cfc54b2>

Potential attributes mentioned in the call:

- Average distance traveled.
- Medical donors.
- Total donors.
- Percent of donor base represented.
- Population numbers.
- Donor reach.
- Possible donor-origin by ZIP information.
- Diversity or demographic information if available.

Implementation guidance:

- Confirm access to the layer in the BioMed group.
- Confirm field names and attribute definitions.
- Confirm whether polygons are red-cell only.
- Confirm whether ZIP-based donor-origin layers should be included or reserved for later.
- Confirm which attributes are appropriate for public or internal display.
- Add the layer to the Data Sources & Methodology modal.
- Use a clear legend explaining what the polygon represents.

Potential explanation:

“Trade areas represent the observed donor reach associated with fixed sites and are based on donor behavior patterns. They help show the market and population context served by each fixed-site location.”

Do not use that final language without confirming the exact methodology with Troy, but the concept should be preserved.

27. Fixed Site Infographic Concept

The call surfaced a strong idea: use the trade-area attributes to generate an infographic for each selected fixed site.

This could become one of the most powerful features in the application because it transforms a GIS layer into an executive narrative.

Instead of showing only a polygon and a table, the application could present a clean panel that says, in effect:

“This fixed site reaches this population, draws donors from this area, and supports this local BioMed footprint.”

Potential infographic components:

- Fixed site name.
- Region.
- Chapter or related geography.
- Trade-area population.
- Total donors.
- Medical donors.
- Average distance traveled.
- Donor-base representation.
- Nearby hospitals served or indirectly supported, if data supports that relationship.
- Diversity or population representation metrics.
- Collection activity.

Design guidance:

- Use summary cards, not a dense table.
- Use plain-language labels.
- Use one sentence explaining what the trade area means.
- Use map and metrics together.
- Let the user return to the region overview easily.

Potential panel heading:

“Fixed Site Market Reach”

Potential explanatory sentence:

“This view summarizes the donor reach and population context associated with the selected fixed-site trade area.”

This feature should be built after the basic Explore Regions structure is stable. It is too important to rush, but it should be kept in the core design.

28. Red Cell and Platelet Trade Area Roadmap

The current fixed-site trade-area discussion focuses on red cell. Troy said the red cell layer is available now and that platelet trade areas may be added later. He also mentioned that platelet areas may be based on a standard radius around fixed sites and that platelet donors may travel farther than red cell donors.

The implementation should therefore be staged.

Phase 1:

- Build using red cell fixed-site trade areas.
- Confirm attributes and methods.
- Build region-to-fixed-site workflow.
- Build selected fixed-site infographic.

Phase 2:

- Add platelet trade areas when Troy provides the layer.
- Decide whether platelet should be a separate layer, toggle, or metric panel.
- Consider whether existing polygons can be reused with platelet metrics as a shortcut.

The call included the idea that the team could “cheat” by using polygons and adding platelet metrics rather than rebuilding everything. That is a practical option, but it should only be used if it is analytically acceptable. The application should not imply that a red-cell polygon represents platelet behavior unless the methodology supports that interpretation.

Recommended design:

- Add a product-type toggle only when both red cell and platelet data are ready.
- Until then, label the current trade areas clearly as red cell fixed-site trade areas.
- Use the Data Sources & Methodology modal to explain the methodology and limitations.

Potential note:

“Current trade-area views reflect red cell fixed-site data. Platelet-specific trade-area views may be added in a later phase.”

29. Sickle Cell Content in Explore Regions

The Sickle Cell content should move to Explore Regions. This was an explicit content realignment request.

The reason is that Sickle Cell is not simply a collection process topic. It is connected to community need, donor diversity, population representation, local engagement, and patient impact. Explore Regions is therefore the proper home.

In Explore Regions, Sickle Cell content can be presented alongside regional demographic context, donor representation, and community impact. This allows users to understand not only that Sickle Cell matters but where it matters and how it connects to local donor strategy.

Implementation guidance:

- Move Sickle Cell content out of BioMed Collections.
- Reframe it as part of community impact.
- Connect it to diversity and population representation where source-supported.
- Avoid unsupported geographic claims.
- Validate language with Marketing & Communications.
- Include sources in the Data Sources & Methodology modal.

Potential section title:

“Sickle Cell and Community Impact”

Potential placeholder language:

“Sickle Cell content should be presented in the regional context because it connects directly to community need, donor diversity, and population representation. Explore Regions should help users understand how this issue appears locally and how BioMed engagement can support community impact.”

30. Data Sources & Methodology Modal

This is a critical requirement and should be part of the application architecture. It should not be a visible tile. It should not be presented as a visible navigation tile. It should be an application-wide modal that users can open when they need source detail, definitions, refresh dates, methodology, or data-owner information.

The reason is simple: the review began with a source question. Troy asked whether there was a list showing where the statistical data came from. He also noted that if someone questions the numbers, the team needs to be able to explain where the data came from and how it was built.

This is not a small documentation issue. It is the credibility layer for the entire application.

Recommended implementation:

- Add a subtle “About the Data” button, link, or information icon.
- Make it available globally or at least on data-heavy tiles.
- Open a modal dialog rather than navigating away from the current page.
- Use accordion sections inside the modal.
- Allow the user to expand only the source area they need.
- Keep users in context so they can close the modal and return to the same map or dashboard.

Recommended modal title:

“Data Sources & Methodology”

Recommended launch labels:

- “About the Data”
- “Data Sources”
- “Methodology”
- Information icon with tooltip: “View data sources and methodology”

The modal should become the place where every major metric has a defensible home.

31. Modal Accordion Structure

The modal should use an accordion layout. This is important because the data-source content could become long. An accordion lets users find what they need without being overwhelmed.

Recommended accordion sections:

Data Sources

This section should list each dataset used in the application.

For each dataset, include:

- Dataset name.
- Application tile or section where it is used.

- Source system or layer.
- Data owner.
- Business owner.
- Technical owner.
- Source link or item ID where appropriate.
- Notes.

Refresh Schedule

This section should explain how current the data is.

For each dataset, include:

- Refresh frequency.
- Last refresh date.
- Next expected refresh date if known.
- Whether refresh is manual or automated.
- Whether data is static, periodic, or near real-time.

Definitions

This section should define terms that may be misunderstood.

Potential definitions:

- Division.
- Region.
- District.
- Chapter.
- Fixed site.
- Mobile collection.
- Trade area.
- Medical donor.
- Products distributed.
- Hospital served.
- Staging site.
- Kitting site.
- IRL site.
- Manufacturing site.
- GOAT hierarchy.

Methodology

This section should explain how calculations and geographies are built.

Potential methodology entries:

- Future Demand calculation method.
- Collection aggregation logic.
- Fixed-site trade-area methodology.
- Regional rollup logic.

- Boundary relationship logic.
- Population representation methodology.
- Product distribution aggregation.

Known Limitations

This section should document assumptions and caveats.

Potential limitations:

- BioMed and Humanitarian Services boundaries may not align exactly.
- Some regions may have no fixed sites.
- Some datasets may refresh on different schedules.
- Some metrics may be national only until regional fields are available.
- Trade-area methodology may differ between red cell and platelet.
- Certain demographic or donor-representation data may require careful interpretation.

Data Steward Contacts

This section should identify who can answer questions.

Potential roles:

- Business owner.
- Technical owner.
- GIS owner.
- Data steward.
- Marketing/Communications reviewer.

This accordion design solves the source-traceability problem without adding clutter to the main experience.

32. Data Source Inventory Requirements

Troy's email includes the instruction to obtain a complete data sources list from Jeff. This should be treated as a formal deliverable.

The source list should include every dataset that supports a statistic, layer, map, dashboard, filter, or chart.

Minimum inventory fields:

- Dataset name.
- Description.
- Source system.
- Source link or item ID.
- Owner.
- Steward.
- Refresh frequency.
- Last refresh date.
- Used in tile.

- Metrics generated from dataset.
- Notes.
- Known limitations.

Initial datasets likely include:

- Fixed Site Map Layer.
- Red Cell Fixed Sites Trade Area Dashboard or source data.
- Collection data.
- Mobile collection data.
- Fixed-site collection data.
- Future Demand data.
- Jurisdiction boundary data.
- GOAT hierarchy data.
- Operations infrastructure data.
- Manufacturing sites.
- Staging sites.
- Kitting sites.
- IRL sites.
- Distribution sites.
- Hospital network data.
- Sickle Cell data.
- Diversity and population representation data.
- Volunteer data.

The source list can be built in NotebookLM if the underlying source material is available. The transcript mentioned that source information had been placed into NotebookLM and that NotebookLM can produce a full document. That output should be used as a starting point, but it should be reviewed and structured into the modal format.

Implementation guidance:

- Do not rely on NotebookLM output without review.
 - Convert the source document into a structured source inventory.
 - Add source inventory entries to the application modal.
 - Add source references to future documentation.
 - Use consistent names for datasets across the application.
-

33. KPI and Metric Standards

The revised application should use a consistent standard for all KPIs and metrics. This is especially important because the experience will likely be reviewed by leadership and may be used beyond the original audience.

Every KPI should answer:

- What does this number represent?
- What is the source?
- What is the time period?

- What geography does it represent?
- Is it a count, rate, percentage, projection, or calculated value?
- How often is it refreshed?
- Who owns the number?
- Are there limitations?

This does not mean every KPI card should display all of that information visibly. The card should remain clean. The details should be available in the Data Sources & Methodology modal.

Recommended KPI card pattern:

- Label.
- Value.
- Short context line.
- Optional info icon.

Example:

Label: "Hospitals Served"

Value: "XX"

Context: "Selected region"

Info icon: Opens modal section for Hospital Network data.

Recommended standard fields for backend documentation:

- KPI name.
- Definition.
- Calculation.
- Source dataset.
- Filter behavior.
- Geographic level.
- Refresh frequency.
- Owner.
- Limitations.

This standard should apply to Future Demand, BioMed Collections, Explore Regions, Hospital Network, Operations Workbench, and any other data-heavy section.

34. Marketing and Communications Review

The email specifically says to validate Blood 101 content with Marketing & Communications. The same standard should apply to other narrative-sensitive content.

Content that should likely be reviewed:

- Volunteer sections.
- Diversity and population representation sections.
- Sickle Cell content.
- Fixed Site Growth Program language.
- Community impact language.

- Any public-facing or executive-facing claims.

The goal of Marketing & Communications review is not to slow down the build. It is to ensure that language is accurate, mission-aligned, and safe for broader distribution.

Implementation guidance:

- Build placeholder sections with clear labels.
- Mark language as draft if not yet approved.
- Avoid hard numbers or claims until sources are confirmed.
- Keep editable text in places that can be easily updated.
- If approval is pending, use neutral language.

Potential internal note to include in working version:

“Narrative language in this section is pending Marketing & Communications validation.”

Do not include that note in the final published application unless stakeholders want it visible.

35. User Experience and Interaction Standards

The revised experience should remain clean and executive-friendly. The application should not feel like a GIS developer tool even though it is built on GIS and dashboard data.

User experience principles:

- Start broad, then let users drill down.
- Do not expose every layer at once.
- Use progressive disclosure.
- Keep labels clear.
- Keep map controls consistent.
- Provide reset buttons where filters can create confusion.
- Use modals and accordions to hide supporting detail until needed.
- Avoid requiring users to understand internal systems before using the app.

Specific interaction recommendations:

- Use dropdowns for Division, Region, and Chapter where data supports them.
- Use bookmarks for GOAT hierarchy navigation in map-heavy tiles.
- Use collapsible side panels in Operations Workbench.
- Use a modal for source detail.
- Use selected-state highlighting for fixed-site trade areas.
- Use cards for metrics.
- Use short explanatory paragraphs between interactive sections.

The experience should feel guided, not cluttered.

36. Implementation Sequence

The build should proceed in phases. This prevents the largest new features from blocking easier improvements.

Phase 1 - Structure and Naming

Complete the structural changes first.

Tasks:

- Change homepage wording to “Inside BioMed Capabilities.”
- Reorder the chapter tiles.
- Merge Biomed Blood Map and Jurisdiction Dashboard into Jurisdiction Dashboard.
- Add Explore Regions as Tile 9.
- Confirm there are exactly nine visible tiles.
- Remove any visible concept of a tenth tile for data sources.
- Establish the global “About the Data” modal access pattern.

Why this phase matters:

The structure is the skeleton of the experience. All later content changes depend on it.

Phase 2 - Content Relocation

Move existing content to its correct narrative home.

Tasks:

- Move Clara AI to Blood Journey.
- Move RapidPass to Blood Journey.
- Move Track Your Blood to Blood Journey.
- Move Sickle Cell to Explore Regions.
- Remove or update old references.
- Check navigation and section titles.

Why this phase matters:

This immediately improves the story without waiting for new data.

Phase 3 - Volunteer and Narrative Additions

Add volunteer visibility and draft narrative sections.

Tasks:

- Add volunteer section to Blood 101.
- Add volunteer section to Blood Journey.
- Add volunteer section to Hospital Distribution.
- Add volunteer section to BioMed Collections.
- Flag language for Marketing & Communications review.

Why this phase matters:

Volunteer visibility was one of the clearest cross-tile requests.

Phase 4 - Data Source and Methodology Framework

Build the credibility layer.

Tasks:

- Create the Data Sources & Methodology modal.
- Add accordion sections.
- Add initial source inventory.
- Add source entries for fixed-site links.
- Add Future Demand source placeholders.
- Add boundary and GOAT hierarchy source placeholders.
- Add definitions.
- Add known limitations.

Why this phase matters:

Source transparency is foundational. It protects every metric in the application.

Phase 5 - BioMed Collections Enhancements

Build the new internal structure.

Tasks:

- Add Mobile Collections section.
- Add Fixed Sites section.
- Add Fixed Site Growth Program content.
- Add Division, Region, and Chapter filters where possible.
- Add territory footprint map visualizations.
- Add diversity and population representation section.
- Add boundary disclaimer.
- Add source-backed metrics.

Why this phase matters:

Collections is a central educational tile and will likely receive close review.

Phase 6 - Operations Workbench Enhancements

Strengthen the already-liked design.

Tasks:

- Retain existing layout.
- Add infrastructure layers.
- Add GOAT hierarchy bookmarks.
- Add Division, Region, District, and Chapter navigation.
- Add collapsible side panel.
- Add boundary disclaimer.
- Confirm real data versus prototype layers.

Why this phase matters:

Operations Workbench is a strong feature. Enhancements should preserve and extend that strength.

Phase 7 - Explore Regions Foundation

Build the regional story framework.

Tasks:

- Create region selector.
- Add regional summary panel.
- Add regional map.
- Add regional KPIs.
- Move and reframe Sickle Cell content.
- Add community impact narrative.
- Add placeholders for metrics Troy will provide.

Why this phase matters:

Explore Regions is the largest new storytelling area and should be built carefully.

Phase 8 - Fixed Site Trade Area Integration

Add the fixed-site drill-down capability.

Tasks:

- Confirm access to FS Map Layer.
- Add red cell fixed-site trade area layer.
- Filter fixed sites by selected region.
- Add fixed-site selector.
- Display selected trade area polygon.
- Add fixed-site infographic panel.
- Add trade-area methodology to modal.
- Add empty-state behavior for regions without fixed sites.

Why this phase matters:

This is likely the most distinctive GIS storytelling feature in the revised experience.

Phase 9 - Hospital Network

Wait for Troy's dashboard concept.

Tasks once received:

- Review dashboard concept.
- Identify data sources.
- Determine whether the tile embeds, references, or rebuilds the dashboard.
- Add hospital network metrics.
- Add source entries to the modal.

Why this phase matters:

Do not overbuild before requirements are clear.

Phase 10 - Validation and Review

Prepare for review with Troy, Jennifer, and other stakeholders.

Tasks:

- Validate Future Demand numbers.
 - Validate source list.
 - Review volunteer language.
 - Review diversity and Sickle Cell language.
 - Test filters.
 - Test boundary disclaimers.
 - Test modal behavior.
 - Test fixed-site drill-down.
 - Confirm all links and layers work.
-

37. Dependencies and Items Needed from Troy

The review identified several items Troy will provide or needs to provide.

Needed items:

- Written notes from the review.
- Complete data source list or NotebookLM source output.
- Confirmation of Future Demand sources.
- Additional KPIs for BioMed Collections.
- Fixed Site Growth Program content.
- Access to the FS Map Layer if not already available.
- Confirmation of fixed-site trade-area attribute definitions.
- Red Cell Fixed Sites Trade Area Dashboard details.
- Platelet layer when ready.
- Hospital Network dashboard concept.
- Distribution dashboard concept or link.
- Additional Explore Regions metrics.
- Marketing & Communications guidance for volunteer, diversity, and Sickle Cell language.

Potential access checks:

- Confirm Jeff has access to the BioMed group.
 - Confirm the FS Map Layer is visible and usable.
 - Confirm dashboard source layers are accessible.
 - Confirm whether any data should be restricted.
-

38. Open Questions

The following questions should be resolved during the next review or as data is provided.

1. What is the final approved source for Future Demand numbers?
2. Which Future Demand numbers surprised Jennifer, and do they require explanation or correction?
3. What KPIs should be added to BioMed Collections?
4. What exact metrics should appear in the Mobile Collections section?
5. What exact metrics should appear in the Fixed Sites section?
6. What language should be used for the Fixed Site Growth Program?
7. Which volunteer metrics, if any, are available and approved?
8. What diversity and population representation metrics are approved for use?
9. Should LGBTQ+ donor information be presented nationally only, or also regionally?
10. What data source supports the Sick Cell section?
11. Should Explore Regions include all regions or only regions with fixed sites?
12. How should the application display a selected region with zero fixed sites?
13. Are red cell trade-area polygons approved for use in this executive experience?
14. What fields in the fixed-site layer should appear in the infographic?
15. When will the platelet trade-area layer be available?
16. Should platelet be a separate view, a toggle, or a later-phase metric addition?
17. What is the authoritative source for GOAT hierarchy boundaries?
18. How should California and other boundary exceptions be documented?
19. What exactly should be included in the Hospital Network tile?
20. Should the Data Sources & Methodology modal be global, tile-specific, or both?

These questions should not block the structural changes. They should guide the next round of data integration and stakeholder review.

39. Quality Assurance Checklist

Before sending the revised experience back for review, the following checks should be completed.

Structure

- Homepage title updated.
- Nine visible tiles only.
- Tiles in correct order.
- Jurisdiction Dashboard merged.
- Explore Regions added.

Content Movement

- Clara AI moved to Blood Journey.
- RapidPass moved to Blood Journey.
- Track Your Blood moved to Blood Journey.
- Sick Cell moved to Explore Regions.
- Old duplicate content removed or cross-referenced.

Volunteer Content

- Volunteer section added to Blood 101.
- Volunteer section added to Blood Journey.
- Volunteer section added to Hospital Distribution.
- Volunteer section added to BioMed Collections.
- Volunteer language flagged or validated with Marketing & Communications.

BioMed Collections

- Mobile Collections section added.
- Fixed Sites section added.
- Fixed Site Growth Program included.
- Diversity section added.
- Boundary disclaimer added.
- Filters implemented where possible.
- Territory maps tested.

Data Governance

- Data Sources & Methodology modal created.
- Accordion sections created.
- Source inventory started.
- Future Demand sources documented or marked pending.
- Fixed-site sources documented.
- Boundary sources documented.
- Known limitations documented.

Operations Workbench

- Existing design retained.
- Infrastructure layers confirmed.
- GOAT bookmarks added.
- Collapsible side panel implemented.
- Boundary disclaimer added.

Explore Regions

- Region selector implemented.
- Regional overview created.
- Regional metrics added or placeholdered.
- Sickle Cell content moved and reframed.
- Fixed-site selector planned or implemented.
- Trade-area layer connected or staged.
- Infographic concept designed.

Review Readiness

- All broken links removed.
- All filters tested.
- All map layers load.
- Source questions have answers or are clearly marked pending.

- The application can be demonstrated cleanly from homepage to Explore Regions.
-

40. Future Project - Real-Time Cancellation Dashboard

The call included a discussion of a future real-time cancellation dashboard. This is not part of the immediate BioMed Experience revision, but it should be documented because it may become an important follow-on project.

The concept is to build a dashboard that uses cancellation data refreshed every 15 minutes. Troy mentioned that the data may come from an ODS server query and that a Python script could pull the query and refresh a feature layer every 15 minutes.

The future dashboard could then incorporate live or forecasted weather data. The goal would be to identify fixed sites or sponsors that may be vulnerable to cancellation because of severe weather or travel disruption. Traffic data was also mentioned as a possible future addition.

Potential components:

- ODS cancellation feed.
- Python refresh script.
- Hosted feature layer updated every 15 minutes.
- Cancellation dashboard.
- Weather overlay.
- Forecast risk zones.
- Traffic overlay.
- Vulnerable fixed-site indicators.
- Vulnerable sponsor indicators.
- Risk scoring.
- Alerting or prioritization.

Potential questions the dashboard could answer:

- Which drives are at elevated cancellation risk?
- Which fixed sites are near severe weather?
- Which sponsors may be affected by traffic or weather?
- Which regions are experiencing cancellation patterns?
- What cancellation events require operational attention?

This future project is separate from the current BioMed Capabilities Experience, but there may be design overlap. The same principles apply: source transparency, clear geography, live data labeling, and user-friendly risk interpretation.

41. Final Interpretation of the Review

The review did not ask for a smaller application. It asked for a more complete one.

The core experience is working. Jennifer liked the design. Troy was enthusiastic. The Operations Workbench was praised. The overall structure has value. The next version should preserve that strength while adding the missing layers of story, credibility, and locality.

The most important changes are not simply tile moves. They are narrative improvements.

Changing the homepage language reframes the application as a capability view.

Reordering the tiles creates a better learning path.

Adding volunteers makes the experience more human and more accurate.

Moving Clara AI, RapidPass, and Track Your Blood into Blood Journey strengthens the donor story.

Moving Sickle Cell into Explore Regions strengthens the community impact story.

Adding Mobile Collections and Fixed Sites gives users a clearer understanding of collection strategy.

Adding the Fixed Site Growth Program connects fixed sites to future strategy.

Adding diversity and population representation gives context to donor engagement and community needs.

Merging the maps reduces redundancy and creates a stronger jurisdiction reference.

Enhancing Operations Workbench preserves a design that already works while making it more useful.

Building Explore Regions gives the application its local heart.

Adding fixed-site trade areas gives the application a distinctive GIS storytelling capability.

Adding the Data Sources & Methodology modal gives the application credibility.

Together, these changes move the experience from a strong prototype into a leadership-ready BioMed capability platform.

The final product should allow a user to understand BioMed at multiple levels: national, regional, operational, community, and fixed-site. It should tell the story of what BioMed does, how it does it, where it operates, who it serves, how volunteers contribute, how communities are represented, and why the data can be trusted.

That is the complete direction of the review.

42. One-Page Builder View

For daily build work, the following condensed view can be used while the larger document remains the source of truth.

1. Change homepage to “Inside BioMed Capabilities.”
2. Build nine visible tiles only.
3. Merge Biomed Blood Map and Jurisdiction Dashboard.
4. Add Explore Regions.
5. Move Clara AI, RapidPass, and Track Your Blood to Blood Journey.
6. Move Sickle Cell to Explore Regions.
7. Add volunteer sections to Blood 101, Blood Journey, Hospital Distribution, and BioMed Collections.
8. Validate Future Demand numbers and sources.

9. Add Mobile Collections section with filters and territory map.
10. Add Fixed Sites section with Fixed Site Growth Program.
11. Add Diversity and population representation section.
12. Add boundary disclaimer wherever BioMed and HS geographies may be compared.
13. Retain Operations Workbench design.
14. Add GOAT hierarchy bookmarks and collapsible panel.
15. Build Explore Regions as regional story, not just a map.
16. Add region selection and fixed-site selection.
17. Use fixed-site trade area layer for red cell drill-down.
18. Build fixed-site infographic panel.
19. Keep Hospital Network pending until Troy sends concept.
20. Implement Data Sources & Methodology as a modal with accordion sections.
21. Use the modal for sources, refresh dates, definitions, methodology, limitations, and data owners.
22. Document all open questions before the next stakeholder review.

This one-page view is not the full specification. It is a field checklist for implementation.

43. Suggested Review Script for the Next Meeting

When presenting the revised experience back to Troy and Jennifer, the demonstration should follow the revised story arc.

Start with the homepage and state that the title has been changed to “Inside BioMed Capabilities” to better reflect the purpose of the experience.

Then show the nine-tile layout and explain that the order now moves from foundation to journey to distribution to demand to collections to geography to operations to hospital network to regional impact.

Show Blood 101 and point out that volunteer visibility has been added as requested.

Show Blood Journey and point out that Clara AI, RapidPass, and Track Your Blood now live where they belong in the donor journey.

Show Hospital Distribution and point out the volunteer addition.

Show Future Demand and explicitly mention that the data is being validated and tied to documented sources.

Show BioMed Collections and explain the new structure: Mobile Collections, Fixed Sites, Fixed Site Growth Program, Diversity, Volunteers, and boundary disclaimer.

Show Jurisdiction Dashboard and explain that the previous map and dashboard concepts have been merged.

Show Operations Workbench and emphasize that the existing design was retained, with added hierarchy navigation and collapsible panel.

Show Hospital Network only if Troy’s concept has been incorporated; otherwise note it remains pending.

Spend the most time on Explore Regions. Explain that it is designed to tell the local story of BioMed. Show region selection, community metrics, Sickle Cell placement, and fixed-site trade-area drill-down if implemented.

Finally, open the Data Sources & Methodology modal. Explain that this was added because the review raised the need to defend every statistic. Show the accordion structure and explain that this keeps the main experience clean while making source detail available when needed.

This review script will help the stakeholders see that every change directly responds to the email and the conversation.

44. Closing Build Principle

The best version of this application is not merely a map experience. It is a credibility system.

It should help leadership see the BioMed mission with clarity. It should help regional users understand their local footprint. It should help planners see infrastructure and boundaries. It should help communicators tell a better story. It should help executives trust the numbers.

That means every design decision should pass this test:

Does this make BioMed easier to understand, easier to trust, or easier to explain?

If the answer is yes, it belongs.

If the answer is no, it should be simplified, moved, or removed.